E14e - Semiconductor Diodes

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1 Introduction

Semiconductor diodes are electronic components with a strongly non-linear current-voltage characteristic. Their behaviour is mainly determined by the properties of the pn-junction, where charge carrier diffusion leads to the formation of a depletion region. This depletion region is responsible for the rectifying behaviour of the diode and also acts as a voltage-dependent junction capacitance. Therefore, both the current-voltage characteristic and the capacitance-voltage characteristic provide important information about the physical properties of semiconductor diodes.

The aim of this experiment was to investigate different types of semiconductor diodes experimentally. First, the current-voltage characteristics $I(U)$ of a silicon diode, a Zener diode, and a red light-emitting diode were measured. The measurements were performed using a linear voltage sweep, and the resulting characteristics were compared in one diagram. From these data, characteristic diode parameters were determined. For the silicon diode, the emission coefficient was obtained from a linear regression of the Shockley equation in a semi-logarithmic representation. For the Zener diode, the breakdown voltage, the direct current resistance, and the differential resistance were determined from the reverse-bias characteristic. For the red LED, the threshold voltage was found by extrapolating the approximately linear part of the forward characteristic, and the corresponding average emitted wavelength was estimated.

In the second part of the experiment, the voltage-dependent junction capacitance of a silicon power diode was measured using a resonance method. The diode was operated under reverse bias in an LCR circuit, and the resonance frequency was determined from the Lissajous figure on the oscilloscope. From the measured resonance frequencies, the total capacitance of the diode was calculated as a function of the applied reverse voltage. A non-linear fit of the capacitance-voltage characteristic was then used to determine the zero-bias depletion-layer capacitance and the corresponding depletion-layer thickness. The experimental results were analysed together with their uncertainties and compared with the expected behaviour of pn-junction diodes.

2 Theoretical Analysis

2.1 Overview

A semiconductor diode consists of a pn-junction, where diffusion of charge carriers leads to the formation of a depletion region. This region produces an internal electric field and causes the diode to conduct current mainly in one direction. In forward bias, the potential barrier is reduced and the current increases strongly with voltage. In reverse bias, only a small leakage current flows until breakdown occurs, as in the case of a Zener diode. The depletion region also behaves like a capacitance, whose value depends on the applied reverse voltage. Therefore, both current-voltage and capacitance-voltage measurements can be used to determine characteristic diode parameters.

2.2 Current-Voltage Characteristic of the Si Diode

For an ideal pn-junction diode, the current-voltage characteristic is described by the Shockley equation,

$$I = I_S \left[\exp\left(\frac{U}{nU_T} \right) - 1 \right], \tag{1}$$

where I_S is the saturation current, U is the diode voltage, n is the emission coefficient, and

$$U_T = \frac{kT}{e} \tag{2}$$

is the thermal voltage. Here, k is the Boltzmann constant, T is the absolute temperature, and e is the elementary charge.

For sufficiently large forward voltages, the current is much larger than the saturation current, $I \gg I_S$. Therefore, the term -1 in Eq. (1) can be neglected:

$$I \approx I_S \exp\left(\frac{U}{nU_T} \right). \tag{3}$$

Taking the natural logarithm gives

$$\ln(I) = \ln(I_S) + \frac{1}{nU_T}U. \tag{4}$$

Thus, a plot of $\ln(I)$ against U should be linear in the appropriate forward-bias region. If the slope of the fitted straight line is m, then

$$m = \frac{1}{nU_T}. \tag{5}$$

The emission coefficient is therefore

$$n = \frac{1}{mU_T}. \tag{6}$$

2.3 Zener Diode Parameters

The Zener diode was investigated in forward and reverse bias. In reverse bias, the current remains small until the breakdown region is reached. The breakdown voltage U_Z can be determined by fitting the approximately linear part of the reverse-bias characteristic and extrapolating it.

The direct current resistance at a certain operating point is defined as

$$R = \frac{U}{I}. \tag{7}$$

It describes the ratio of voltage to current at that point. The differential resistance is defined by

$$r_d = \frac{dU}{dI}. \tag{8}$$

It describes the local slope of the $U(I)$ curve. If the breakdown region is approximated by a linear fit,

$$U(I) = aI + b, \tag{9}$$

then the differential resistance is directly given by

$$r_d = a. \tag{10}$$

In this experiment, R and r_d are evaluated at

$$I = 0.2I_{Z,\max}. \tag{11}$$

2.4 LED Threshold Voltage and Emitted Wavelength

For a light-emitting diode, charge carriers recombine and emit photons. The energy gained by a charge carrier when passing through the threshold voltage U_S is approximately

$$E = eU_S. \tag{12}$$

The energy of a photon is

$$E = hf = \frac{hc}{\lambda}, \tag{13}$$

where h is Planck's constant, c is the speed of light, and λ is the wavelength. Equating both expressions gives

$$eU_S = \frac{hc}{\lambda}. \tag{14}$$

Solving for the wavelength gives

$$\lambda = \frac{hc}{eU_S}. \tag{15}$$

The threshold voltage U_S is obtained by fitting the approximately linear part of the forward-bias LED characteristic. If the fit is written as

$$I(U) = aU + b, \tag{16}$$

then the threshold voltage is the voltage-axis intercept, where $I = 0$:

$$0 = aU_S + b. \tag{17}$$

Therefore,

$$U_S = -\frac{b}{a}. \tag{18}$$

2.5 Resonance Method for the Junction Capacitance

The junction capacitance of the Si power diode was determined using a resonance method. The diode capacitance forms an oscillating circuit with the inductance L. The resonance frequency of an ideal LC circuit is

$$f = \frac{1}{2\pi\sqrt{LC_{\text{tot}}}}, \tag{19}$$

where C_{tot} is the total measured capacitance. Solving for C_{tot} gives

$$C_{\text{tot}} = \frac{1}{(2\pi f)^2 L}. \tag{20}$$

Thus, the capacitance can be calculated from the measured resonance frequency.

The total capacitance consists of the voltage-dependent depletion-layer capacitance C_S and an additional voltage-independent parallel capacitance C_G:

$$C_{\text{tot}} = C_S + C_G. \tag{21}$$

For a pn-junction, the depletion-layer capacitance decreases with increasing reverse-bias voltage. The fit function used for the measured capacitance is

$$C_{\text{tot}}(U) = C_G + C_0\sqrt{\frac{U_D}{U_D - U}}, \tag{22}$$

where C_0 is the depletion-layer capacitance at zero applied voltage and U_D is the diffusion voltage. Since the diode was connected in reverse direction, the applied voltage in the analysis is taken as negative.

2.6 Depletion-Layer Thickness

The depletion layer can be approximated as a parallel-plate capacitor. Therefore,

$$C_S = \frac{\varepsilon_r \varepsilon_0 A}{d_S}, \tag{23}$$

where ε_r is the relative permittivity of silicon, ε_0 is the vacuum permittivity, A is the junction area, and d_S is the depletion-layer thickness. Solving for d_S gives

$$d_S = \frac{\varepsilon_r \varepsilon_0 A}{C_S}. \tag{24}$$

Since

$$C_S = C_{\text{tot}} - C_G, \tag{25}$$

the depletion-layer thickness as a function of applied voltage is

$$d_S(U) = \frac{\varepsilon_r \varepsilon_0 A}{C_{\text{tot}}(U) - C_G}. \tag{26}$$

For vanishing applied voltage, $C_S = C_0$, so the zero-bias depletion-layer thickness is

$$d_0 = \frac{\varepsilon_r \varepsilon_0 A}{C_0}. \tag{27}$$

3 Experimental Analysis

3.1 Experimental Setup

The experiment consisted of two main experimental setups. The first setup was used to measure the current-voltage characteristics of the Si diode, the Zener diode, and the red LED. The second setup was used to determine the voltage-dependent junction capacitance of a Si power diode by means of a resonance method.

For the measurement of the current-voltage characteristics, the diode under investigation was connected in series with a protective shunt resistor. In contrast to the value suggested in the setup guide, a shunt resistance of

$$R_{\text{shunt}} = 1000\,\Omega \tag{28}$$

was used. The voltage drop across the diode and the current through the diode were measured using the computer-controlled measurement system. The upper multimeter was used for the voltage measurement, while the lower multimeter was used for the current measurement. The DC voltage was varied automatically by the software using a linear voltage sweep.

The Si diode and the red LED were measured only in forward bias. The Zener diode was measured in both forward and reverse bias. For the reverse-bias measurement of the Zener diode, the measured values were later plotted with negative voltage and negative current values in order to represent the reverse branch correctly in the $I(U)$ diagram.

For the capacitance measurement in Task 3, a Si power diode was connected in reverse direction in an LCR series circuit. The circuit contained the diode capacitance, a coil with

$$L = 220\,\mathrm{mH}, \tag{29}$$

a resistor

$$R_M = 100\,\Omega, \tag{30}$$

and a protective resistor

$$R_V = 1000\,\Omega. \tag{31}$$

A capacitor with

$$C = 2\,\mu\mathrm{F} \tag{32}$$

was also used according to the circuit given in the experiment manual. A small sinusoidal AC voltage was supplied by the function generator, while a DC voltage from the power supply was superimposed in order to change the reverse bias of the diode. Although positive voltages were set on the DC supply, the diode was mounted in reverse direction; therefore, the applied diode voltage was treated as negative in the analysis:

$$U = -U_{\mathrm{supply}}. \tag{33}$$

The resonance condition was determined using the oscilloscope in xy-mode. At resonance, the Lissajous figure becomes approximately a straight line. The corresponding resonance frequency was then recorded for each applied reverse-bias voltage.

3.2 Data Acquisition and Measurement Procedure

For Task 1, the current-voltage characteristics were recorded using the measurement software in linear voltage-control mode. Four measurements were performed: Si diode forward bias, Zener diode forward bias, Zener diode reverse bias, and red LED forward bias. For all four measurements, the start voltage was

$$U_{\mathrm{start}} = 1.0 \times 10^{-3}\,\mathrm{V}. \tag{34}$$

The end voltage entered in the software was not used as the final measurement value, since each measurement was stopped manually once enough data points had been collected for the required analysis.

The voltage step size for the Si diode measurement was

$$\Delta U_{\mathrm{Si}} = 1.0 \times 10^{-1}\,\mathrm{V}. \tag{35}$$

For the Zener diode forward measurement, the Zener diode reverse measurement, and the red LED measurement, the voltage step size was

$$\Delta U = 2.0 \times 10^{-1}\,\mathrm{V}. \tag{36}$$

The maximum current limit was set to

$$I_{\mathrm{max}} = 2.0 \times 10^{-2}\,\mathrm{A} \tag{37}$$

for the Si diode, Zener diode forward-bias measurement, and red LED. For the Zener diode reverse-bias measurement, the maximum current was limited to

$$I_{\mathrm{max,Z}} = 1.0 \times 10^{-2}\,\mathrm{A}. \tag{38}$$

The measured data from Task 1 were then used for the analysis in Task 2. The Si diode data were analysed using a semi-logarithmic plot of $\ln(I)$ against U, from which the emission coefficient was determined by linear regression in the approximately linear region. The Zener diode reverse-bias data were used to determine the breakdown voltage, as well as the direct current resistance and differential resistance at $0.2I_{Z,\mathrm{max}}$. The red LED data were fitted in the approximately linear forward-bias region in order to determine the threshold voltage. From this threshold voltage, the average emitted wavelength was calculated.

For Task 3, the DC supply voltage was varied over 15 different values. Since the diode was connected in reverse direction, these values correspond to reverse-bias voltages in the range

$$U = -0.5\,\mathrm{V} \quad \text{to} \quad U = -30\,\mathrm{V}. \tag{39}$$

For each DC voltage, the frequency of the AC signal was adjusted until the Lissajous figure on the oscilloscope became approximately a straight line. This frequency was recorded as the resonance frequency. The total capacitance of the diode was then calculated from the resonance condition,

$$C_{\mathrm{tot}} = \frac{1}{(2\pi f)^2 L}. \tag{40}$$

The capacitance data were fitted using the voltage-dependent capacitance model

$$C_{\mathrm{tot}}(U) = C_G + C_0\sqrt{\frac{U_D}{U_D - U}}. \tag{41}$$

From the fitted value of C_0, the depletion-layer thickness at zero applied voltage was calculated. The voltage-dependent depletion-layer thickness was also determined using

$$d_S(U) = \frac{\varepsilon_r \varepsilon_0 A}{C_{\mathrm{tot}}(U) - C_G}. \tag{42}$$

Since exact instrument uncertainties were not available, estimated uncertainties were used. For the Task 1 voltage and current measurements, the uncertainties were taken as

$$\Delta U = 0.005\,\mathrm{V} \tag{43}$$

and

$$\Delta I = 0.05\,\mathrm{mA}. \tag{44}$$

For the resonance measurement in Task 3, the frequency uncertainty was estimated as

$$\Delta f = 0.05\,\mathrm{kHz}. \tag{45}$$

The inductance uncertainty was treated separately as a systematic uncertainty. A relative tolerance of

$$\frac{\Delta L}{L} = 5\% \tag{46}$$

was assumed for the coil. This systematic uncertainty affects the absolute capacitance scale and therefore also the calculated depletion-layer thickness, but it does not significantly affect the relative voltage dependence of the capacitance.

4 Current-Voltage Characteristics (Task 1)

4.1 Goal

The goal of the first task was to measure the current-voltage characteristics $I(U)$ of three different semiconductor diodes: a silicon diode, a Zener diode, and a red light-emitting diode. The Si diode and the red LED were measured in forward bias only, while the Zener diode was measured in both forward and reverse bias. The obtained characteristics were then plotted together in one diagram in order to compare their behaviour.

4.2 Measurement Method

The measurements were performed using the computer-controlled measurement software in linear voltage-control mode. The current through the diode and the voltage across the diode were recorded for each measurement point. A protective shunt resistor with

$$R_{\mathrm{shunt}} = 1000\,\Omega \tag{47}$$

was connected in series with the diode.

For all four measurements, the voltage sweep was started at

$$U_{\mathrm{start}} = 1.0 \times 10^{-3}\,\mathrm{V}. \tag{48}$$

The measurements were stopped manually once a sufficient number of data points had been collected. Therefore, the final voltage values do not necessarily correspond exactly to the end voltages entered in the software.

For the Si diode, the voltage step size was

$$\Delta U = 1.0 \times 10^{-1}\,\mathrm{V}. \tag{49}$$

For the Zener diode in forward bias, the Zener diode in reverse bias, and the red LED, the voltage step size was

$$\Delta U = 2.0 \times 10^{-1}\,\mathrm{V}. \tag{50}$$

The maximum current limit was set to

$$I_{\mathrm{max}} = 2.0 \times 10^{-2}\,\mathrm{A} \tag{51}$$

for the Si diode, the Zener diode in forward bias, and the red LED. For the Zener diode in reverse bias, the maximum current was limited to

$$I_{\mathrm{max,Z}} = 1.0 \times 10^{-2}\,\mathrm{A}. \tag{52}$$

The Zener reverse-bias measurement was recorded with positive voltage and current values by the software. Since the diode was physically operated in reverse bias, these values were represented as negative values in the combined $I(U)$ diagram:

$$U_{\mathrm{reverse}} = -U_{\mathrm{measured}}, \tag{53}$$

$$I_{\mathrm{reverse}} = -I_{\mathrm{measured}}. \tag{54}$$

4.3 Results

The measured current-voltage characteristics are shown in Fig. 1. The current is plotted as a function of the diode voltage, following the convention $I(U)$. Estimated measurement uncertainties of

$$\Delta U = 0.005\,\mathrm{V} \tag{55}$$

and

$$\Delta I = 0.05\,\mathrm{mA} \tag{56}$$

were used for the error bars.

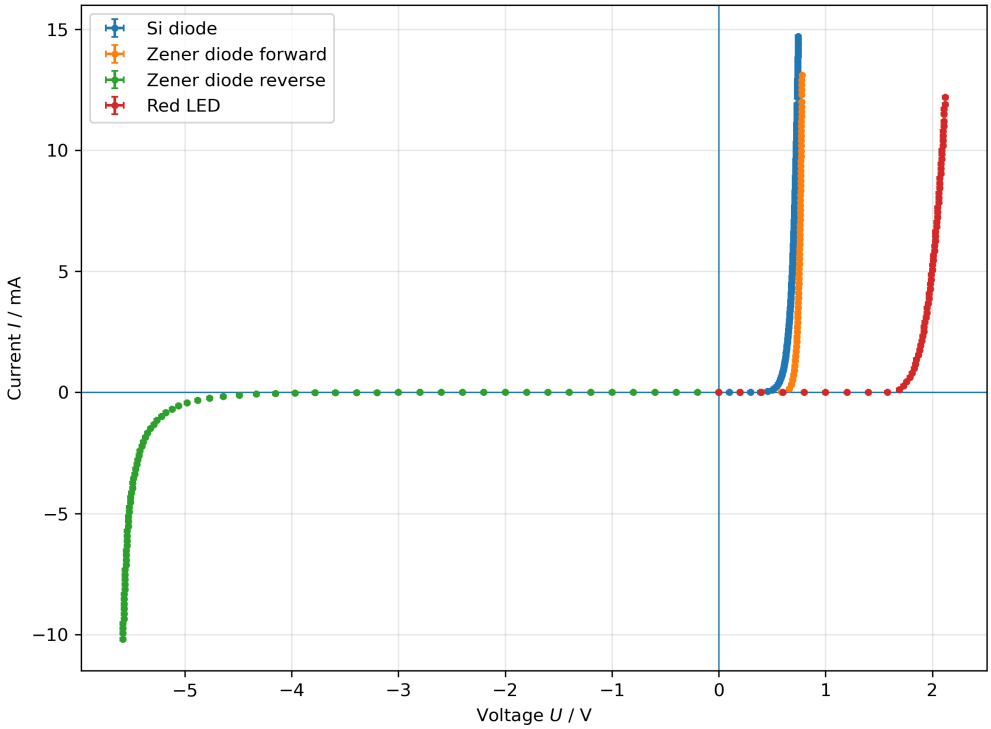


Figure 1: Measured current-voltage characteristics $I(U)$ of the Si diode, the Zener diode in forward and reverse bias, and the red LED.

The Si diode shows the expected non-linear forward characteristic. At low forward voltages, only a very small current flows. As the applied voltage increases, the current rises rapidly, which is typical for a pn-junction diode. The Zener diode in forward bias behaves similarly to the Si diode, since it also acts like a normal diode in the forward direction.

In reverse bias, the Zener diode initially allows only a small current. However, once the reverse voltage becomes sufficiently large, the current increases strongly. This corresponds to the breakdown region of the Zener diode. The reverse branch is therefore clearly different from the forward-bias behaviour.

The red LED also shows a non-linear forward characteristic, but its current starts to increase noticeably only at a higher voltage than for the Si diode. This is expected because the threshold voltage of an LED is related to the photon energy of the emitted light. Since a red LED was used, the threshold voltage should be lower than for LEDs with shorter-wavelength emission, such as blue LEDs, but still larger than that of a normal Si diode.

Overall, the measured characteristics agree qualitatively with the expected behaviour of the different diode types. The data from this task are used in Task 2 to determine the emission coefficient of the Si diode, the breakdown voltage and resistances of the Zener diode, and the threshold voltage and average emitted wavelength of the red LED.

5 Determination of Diode Parameters (Task 2)

5.1 Goal

The goal of Task 2 was to determine characteristic diode parameters from the current-voltage data measured in Task 1. For the Si diode, the emission coefficient n was obtained from a linear regression of the Shockley equation in a semi-logarithmic representation. For the Zener diode, the breakdown voltage U_Z, the direct current resistance R, and the differential resistance r_d were determined from the reverse-bias characteristic. For the red LED, the threshold voltage U_S was found by extrapolating the approximately linear part of the forward characteristic, and the corresponding average emitted wavelength was calculated.

5.2 Analysis Method

5.2.1 Si Diode

The forward characteristic of the Si diode was analysed using the Shockley equation,

$$I = I_S \left[\exp\left(\frac{U}{nU_T} \right) - 1 \right], \tag{57}$$

where I_S is the saturation current, n is the emission coefficient, and U_T is the thermal voltage. The thermal voltage is given by

$$U_T = \frac{kT}{e}. \tag{58}$$

Room temperature was assumed,

$$T = 293.15\,\mathrm{K}, \tag{59}$$

which gives

$$U_T = 25.262\,\mathrm{mV}. \tag{60}$$

For sufficiently large forward currents, $I \gg I_S$, the Shockley equation can be approximated by

$$I \approx I_S \exp\left(\frac{U}{nU_T} \right). \tag{61}$$

Taking the natural logarithm gives

$$\ln(I) = \ln(I_S) + \frac{1}{nU_T} U. \tag{62}$$

Therefore, the emission coefficient can be calculated from the slope m of the linear fit in the $\ln(I)$-U diagram:

$$n = \frac{1}{mU_T}. \tag{63}$$

The fit was performed in the interval

$$0.005\,\mathrm{mA} < I < 2.0\,\mathrm{mA}. \tag{64}$$

This range was chosen because it is approximately linear in the semi-logarithmic representation and avoids the low-current noise region as well as stronger high-current series resistance effects.

5.2.2 Zener Diode

For the Zener diode, the reverse-bias characteristic was used to determine the breakdown behaviour. The analysis was performed using the magnitudes of voltage and current. The approximately linear breakdown region was fitted with

$$|U| = a|I| + b. \tag{65}$$

The breakdown voltage was determined from the intercept of this fit:

$$U_Z = b. \tag{66}$$

The differential resistance is the local slope of the $U(I)$ characteristic,

$$r_d = \frac{dU}{dI}. \tag{67}$$

For the linear fit, this corresponds to

$$r_d = a. \tag{68}$$

The direct current resistance was calculated at the specified current

$$I = 0.2 I_{Z,\max} \tag{69}$$

using

$$R = \frac{U}{I}. \tag{70}$$

The maximum measured reverse current was

$$I_{Z,\max} = 10.200\,\mathrm{mA}, \tag{71}$$

so the evaluation current was

$$0.2 I_{Z,\max} = 2.040\,\mathrm{mA}. \tag{72}$$

The fit of the breakdown region was carried out for

$$|I| > 5.0\,\mathrm{mA}. \tag{73}$$

5.2.3 Red LED

For the red LED, the threshold voltage U_S was determined by fitting the approximately linear high-current region of the forward characteristic. The fit function was

$$I = aU + b. \tag{74}$$

The threshold voltage was obtained from the voltage-axis intercept, where $I = 0$:

$$U_S = -\frac{b}{a}. \tag{75}$$

The fit was performed for

$$I > 8.0\,\mathrm{mA}. \tag{76}$$

The average emitted wavelength was estimated by assuming that the electrical energy eU_S is converted into photon energy:

$$eU_S = \frac{hc}{\lambda}. \tag{77}$$

Thus,

$$\lambda = \frac{hc}{eU_S}. \tag{78}$$

5.3 Results

5.3.1 Si Diode

The semi-logarithmic plot of the Si diode characteristic is shown in Fig. 2. The orange points mark the region used for the linear regression. The fit follows the selected points well, confirming that this region is suitable for applying the linearized Shockley equation.

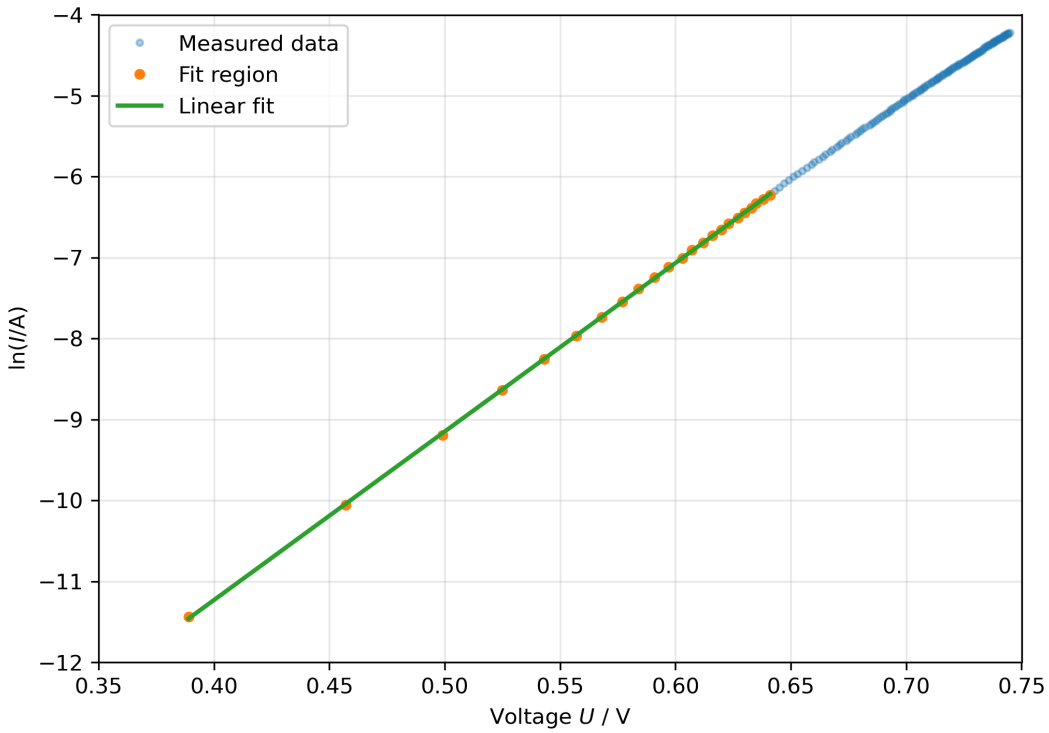


Figure 2: Semi-logarithmic representation of the Si diode forward characteristic.

The fitted slope was

$$m = (2.0821 \pm 0.0038) \times 10^{1}\,\mathrm{V}^{-1}. \tag{79}$$

Using $U_T = 25.262\,\mathrm{mV}$, the emission coefficient was found to be

$$n = 1.901 \pm 0.003. \tag{80}$$

5.3.2 Zener Diode

The reverse-bias breakdown region of the Zener diode is shown in Fig. 3. The data are plotted using the magnitudes of current and voltage. The selected fit region corresponds to the approximately linear part of the breakdown characteristic. The vertical dashed line marks the current $0.2I_{Z,\max}$, and the horizontal dotted line marks the corresponding voltage magnitude.

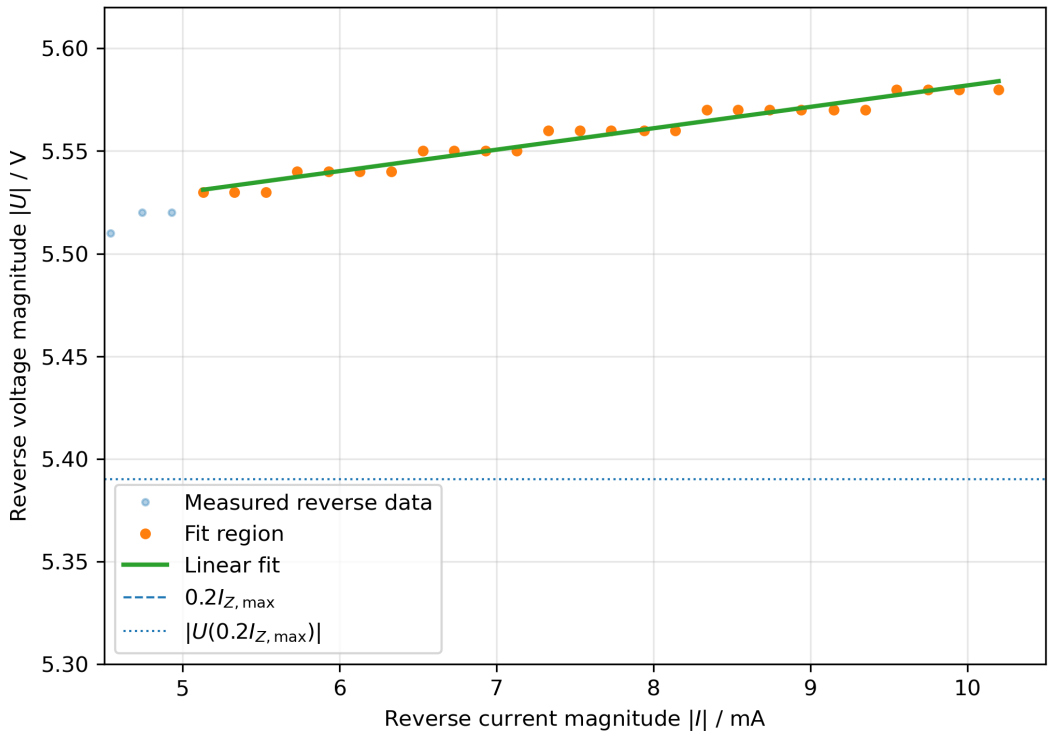


Figure 3: Linear fit to the reverse-bias breakdown region of the Zener diode.

The measured maximum reverse current was

$$I_{Z,\max} = 10.200\,\mathrm{mA}, \tag{81}$$

so that

$$0.2I_{Z,\max} = 2.040\,\mathrm{mA}. \tag{82}$$

At this current, the reverse voltage magnitude was

$$|U(0.2I_{Z,\max})| = 5.390\,\mathrm{V}. \tag{83}$$

The breakdown voltage obtained from the linear extrapolation was

$$U_Z = 5.478 \pm 0.003\,\mathrm{V}. \tag{84}$$

The direct current resistance was

$$R = 2642.2 \pm 64.8\,\Omega, \tag{85}$$

and the differential resistance was

$$r_d = 10.4 \pm 0.4\,\Omega. \tag{86}$$

5.3.3 Red LED

The threshold-voltage fit for the red LED is shown in Fig. 4. The orange points represent the high-current region used for the linear regression. The threshold voltage was obtained by extrapolating this linear fit to $I = 0$.

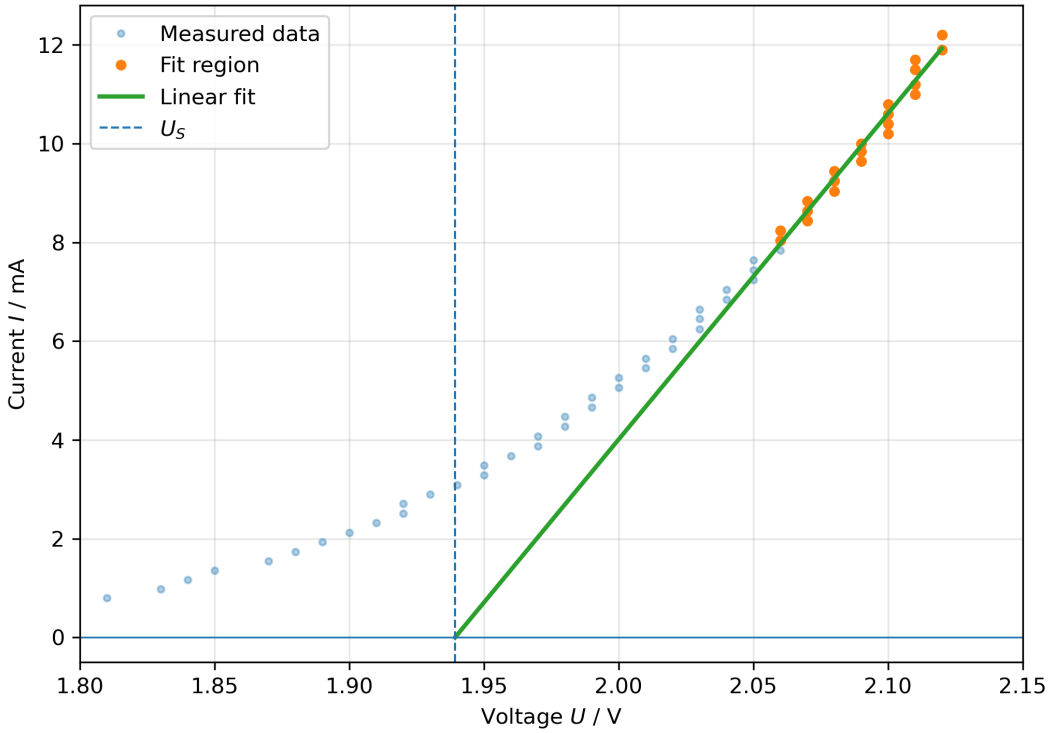


Figure 4: Forward characteristic of the red LED with a linear fit to the high-current region.

The linear fit gave

$$a = (6.599 \pm 0.272) \times 10^{1}\,\mathrm{mA\,V^{-1}}, \tag{87}$$

and

$$b = -127.97 \pm 5.69\,\mathrm{mA}. \tag{88}$$

The resulting threshold voltage was

$$U_S = 1.939 \pm 0.006\,\mathrm{V}. \tag{89}$$

The corresponding average emitted wavelength was

$$\lambda = 639 \pm 2\,\mathrm{nm}. \tag{90}$$

5.4 Summary of Results

The results of the Task 2 analysis are summarized in Table 1.

Table 1: Summary of the diode parameters determined in Task 2.

Quantity	Result
Thermal voltage U_T	25.262 mV
Si diode emission coefficient n	1.901 ± 0.003
Zener maximum reverse current $I_{Z,\max}$	10.200 mA
Evaluation current $0.2I_{Z,\max}$	2.040 mA
Zener voltage at $0.2I_{Z,\max}$	5.390 V
Zener breakdown voltage U_Z	5.478 ± 0.003 V
Direct current resistance R	$2642.2 \pm 64.8\,\Omega$
Differential resistance r_d	$10.4 \pm 0.4\,\Omega$
LED threshold voltage U_S	1.939 ± 0.006 V
LED average wavelength λ	639 ± 2 nm

5.5 Uncertainty Analysis

Since exact instrument uncertainties were not available, estimated measurement uncertainties were used:

$$\Delta U = 0.005\,\mathrm{V}, \tag{91}$$

and

$$\Delta I = 0.05\,\mathrm{mA}. \tag{92}$$

The uncertainties of the fit parameters were obtained from the covariance matrices of the linear regressions.

For the Si diode, the emission coefficient is calculated from

$$n = \frac{1}{mU_T}. \tag{93}$$

Since U_T was calculated from the assumed room temperature and treated as constant, the uncertainty in n was obtained from the uncertainty of the slope m:

$$\Delta n = n\frac{\Delta m}{m}. \tag{94}$$

This gives

$$\Delta n = 0.003. \tag{95}$$

For the Zener diode, the uncertainty of the breakdown voltage was obtained from the uncertainty of the intercept of the linear fit. The uncertainty of the differential resistance was obtained from the uncertainty of the slope:

$$\Delta r_d = \Delta a. \tag{96}$$

The uncertainty of the direct current resistance was calculated using standard error propagation for

$$R = \frac{U}{I}. \tag{97}$$

This gives

$$\Delta R = R\sqrt{\left(\frac{\Delta U}{U}\right)^2 + \left(\frac{\Delta I}{I}\right)^2}. \tag{98}$$

For the LED, the threshold voltage was calculated from

$$U_S = -\frac{b}{a}. \tag{99}$$

The uncertainty in U_S was obtained from the uncertainties and covariance of the fit parameters a and b:

$$(\Delta U_S)^2 = \left(\frac{\partial U_S}{\partial a}\Delta a\right)^2 + \left(\frac{\partial U_S}{\partial b}\Delta b\right)^2 + 2\frac{\partial U_S}{\partial a}\frac{\partial U_S}{\partial b}\mathrm{cov}(a,b). \tag{100}$$

The partial derivatives are

$$\frac{\partial U_S}{\partial a} = \frac{b}{a^2}, \tag{101}$$

and

$$\frac{\partial U_S}{\partial b} = -\frac{1}{a}. \tag{102}$$

The wavelength uncertainty follows from

$$\lambda = \frac{hc}{eU_S}. \tag{103}$$

Therefore,

$$\Delta\lambda = \lambda\frac{\Delta U_S}{U_S}. \tag{104}$$

The numerical uncertainties quoted in Table 1 are therefore mainly determined by the regression uncertainties and the estimated current and voltage measurement uncertainties.

5.6 Discussion

The Si diode shows a clear linear region in the $\ln(I)$-U representation. The obtained emission coefficient

$$n = 1.901 \pm 0.003 \tag{105}$$

is close to the typical range expected for a real Si diode. A value larger than one indicates non-ideal behaviour, which may be caused by recombination effects in the depletion region, series resistance, or deviations from the ideal Shockley model. The selected fit interval avoids the very low-current region, where measurement noise becomes more relevant, and the high-current region, where series resistance effects can distort the exponential characteristic.

For the Zener diode, the reverse characteristic shows a clear breakdown region. The measured breakdown voltage was

$$U_Z = 5.478 \pm 0.003\,\mathrm{V}. \tag{106}$$

The exact nominal Zener diode type was not known, so the result cannot be assigned to a specific nominal value. However, the obtained value is of the expected order of magnitude for a low-voltage Zener diode. The direct current resistance at $0.2I_{Z,\max}$ was much larger than the differential resistance. This is expected because the direct current resistance describes the ratio U/I at a specific operating point, while the differential resistance describes only the local slope in the breakdown region. Since the reverse voltage changes only weakly with increasing current in the breakdown region, the differential resistance is comparatively small.

For the red LED, the threshold voltage was determined as

$$U_S = 1.939 \pm 0.006\,\mathrm{V}. \tag{107}$$

The resulting wavelength was

$$\lambda = 639 \pm 2\,\mathrm{nm}. \tag{108}$$

This wavelength lies in the red region of the visible spectrum and is therefore consistent with the observed colour of the LED. However, the method is only an approximation because an LED does not emit at a single wavelength, and the threshold voltage is not exactly identical to the photon energy. In addition, the fitted threshold voltage depends on the chosen high-current fit region.

Overall, the results agree well with the expected behaviour of the three diode types. The Si diode follows the Shockley equation in an appropriate forward-bias region, the Zener diode shows reverse breakdown, and the red LED gives a threshold voltage corresponding to red light emission.

6 Junction Capacitance of a Si Power Diode (Task 3)

6.1 Goal

The goal of Task 3 was to determine the voltage-dependent junction capacitance of a Si power diode using a resonance method. The diode was operated under reverse bias in an LCR circuit, and the resonance frequency was measured for different applied DC voltages. From these frequencies, the total capacitance of the diode was calculated. A non-linear fit of the capacitance-voltage characteristic was then used to determine the zero-bias depletion-layer capacitance C_0, the voltage-independent parallel capacitance C_G, the diffusion voltage U_D, and the depletion-layer thickness for vanishing reverse bias d_0.

6.2 Measurement Method

The Si power diode was connected in reverse direction in an LCR series circuit. The circuit contained an inductance

$$L = 220\,\mathrm{mH}, \tag{109}$$

a resistor

$$R_M = 100\,\Omega, \tag{110}$$

and a protective resistor

$$R_V = 1000\,\Omega. \tag{111}$$

A small AC voltage was applied using the function generator, and a DC bias voltage was superimposed using the DC power supply. Although positive voltages were set on the power supply, the diode was mounted in reverse direction. Therefore, the voltage used in the analysis was taken as

$$U = -U_{\mathrm{supply}}. \tag{112}$$

For each applied reverse-bias voltage, the frequency was adjusted until the Lissajous figure on the oscilloscope became approximately a straight line. This condition corresponds to resonance. In total, 15 measurements were taken in the voltage range

$$U = -0.5\,\mathrm{V} \quad \text{to} \quad U = -30\,\mathrm{V}. \tag{113}$$

6.3 Capacitance Calculation

At resonance, the total capacitance of the diode was calculated using the Thomson resonance condition,

$$f = \frac{1}{2\pi\sqrt{LC_{\mathrm{tot}}}}. \tag{114}$$

Solving for the capacitance gives

$$C_{\mathrm{tot}} = \frac{1}{(2\pi f)^2 L}. \tag{115}$$

The total measured capacitance contains the voltage-dependent depletion-layer capacitance C_S and an additional voltage-independent parallel capacitance C_G:

$$C_{\mathrm{tot}} = C_S + C_G. \tag{116}$$

The measured capacitance was fitted using the model

$$C_{\mathrm{tot}}(U) = C_G + C_0 \sqrt{\frac{U_D}{U_D - U}}, \tag{117}$$

where C_0 is the depletion-layer capacitance at zero applied voltage and U_D is the diffusion voltage.

6.4 Measured Data

The measured resonance frequencies and the calculated capacitances are listed in Table 2. The capacitance decreases with increasing magnitude of the reverse-bias voltage, as expected for a pn-junction depletion capacitance.

Table 2: Measured resonance frequencies and calculated total capacitances for the Si power diode.

U / V	f / kHz	C_{tot} / nF	C_S / nF	d_S / μm
−0.5	9.8389	1.1894	1.1186	2.3350
−1.0	10.7179	1.0023	0.9315	2.8040
−2.0	11.9629	0.8045	0.7338	3.5597
−3.0	12.8382	0.6986	0.6278	4.1606
−5.0	14.0736	0.5813	0.5105	5.1162
−7.0	14.9786	0.5132	0.4424	5.9040
−10.0	16.0286	0.4482	0.3774	6.9215
−12.0	16.6026	0.4177	0.3469	7.5290
−15.0	17.3417	0.3829	0.3121	8.3697
−18.0	17.9757	0.3563	0.2855	9.1473
−20.0	18.3557	0.3417	0.2709	9.6403
−22.0	18.7087	0.3290	0.2582	10.1172
−25.0	19.1917	0.3126	0.2418	10.8012
−27.0	19.4897	0.3031	0.2323	11.2423
−30.0	19.9067	0.2905	0.2198	11.8851

6.5 Capacitance-Voltage Fit

The calculated total capacitance C_{tot} as a function of reverse-bias voltage is shown in Fig. 5. The non-linear fit was performed using Eq. (117).

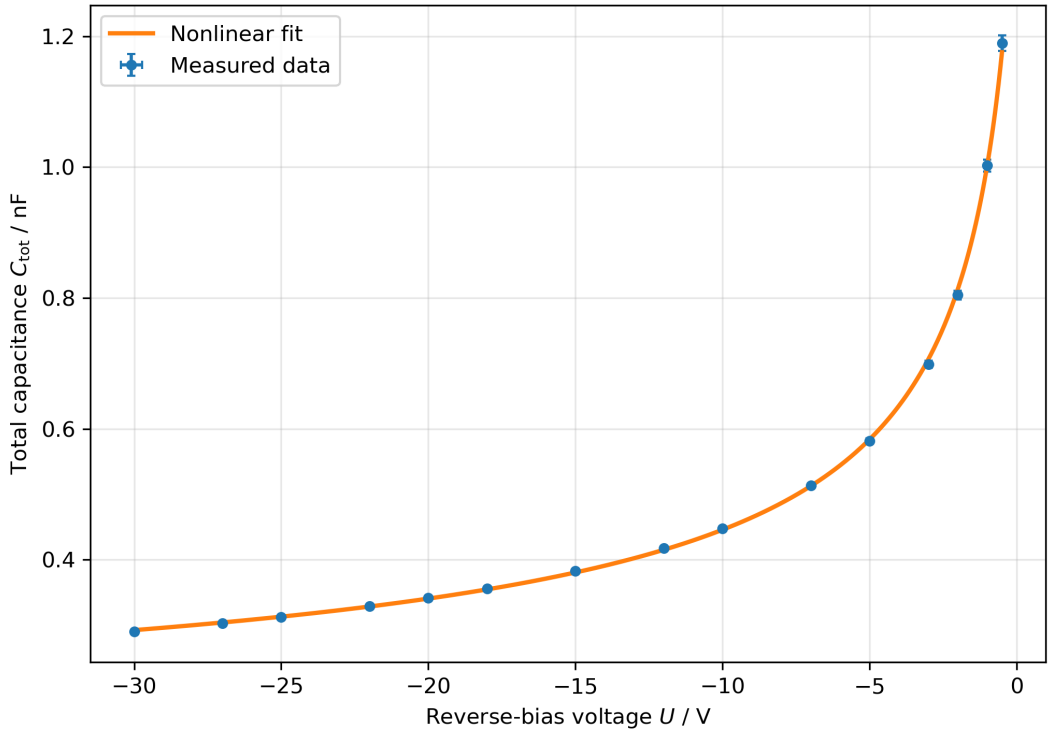


Figure 5: Total capacitance C_{tot} of the Si power diode as a function of reverse-bias voltage.

The fit parameters were obtained as

$$C_G = 0.0708 \pm 0.0031\,\mathrm{nF}, \tag{118}$$

$$C_0 = 1.4306 \pm 0.0242\,\mathrm{nF}, \tag{119}$$

and

$$U_D = 0.7382 \pm 0.0388\,\mathrm{V}. \tag{120}$$

The reduced chi-square value was

$$\chi^2_{\mathrm{red}} = 0.975. \tag{121}$$

Since this value is close to 1, the chosen fit model describes the measured capacitance data well within the assumed frequency uncertainty.

6.6 Depletion-Layer Thickness

The depletion-layer capacitance was calculated by subtracting the voltage-independent contribution from the total capacitance:

$$C_S(U) = C_{\mathrm{tot}}(U) - C_G. \tag{122}$$

The depletion layer was then approximated as a parallel-plate capacitor,

$$C_S = \frac{\varepsilon_r \varepsilon_0 A}{d_S}. \tag{123}$$

Solving for the depletion-layer thickness gives

$$d_S(U) = \frac{\varepsilon_r \varepsilon_0 A}{C_S(U)}. \tag{124}$$

The values

$$\varepsilon_r = 11.8 \tag{125}$$

and

$$A = 25\,\mathrm{mm}^2 \tag{126}$$

were used for the Si power diode. The resulting voltage-dependent depletion-layer thickness is shown in Fig. 6.

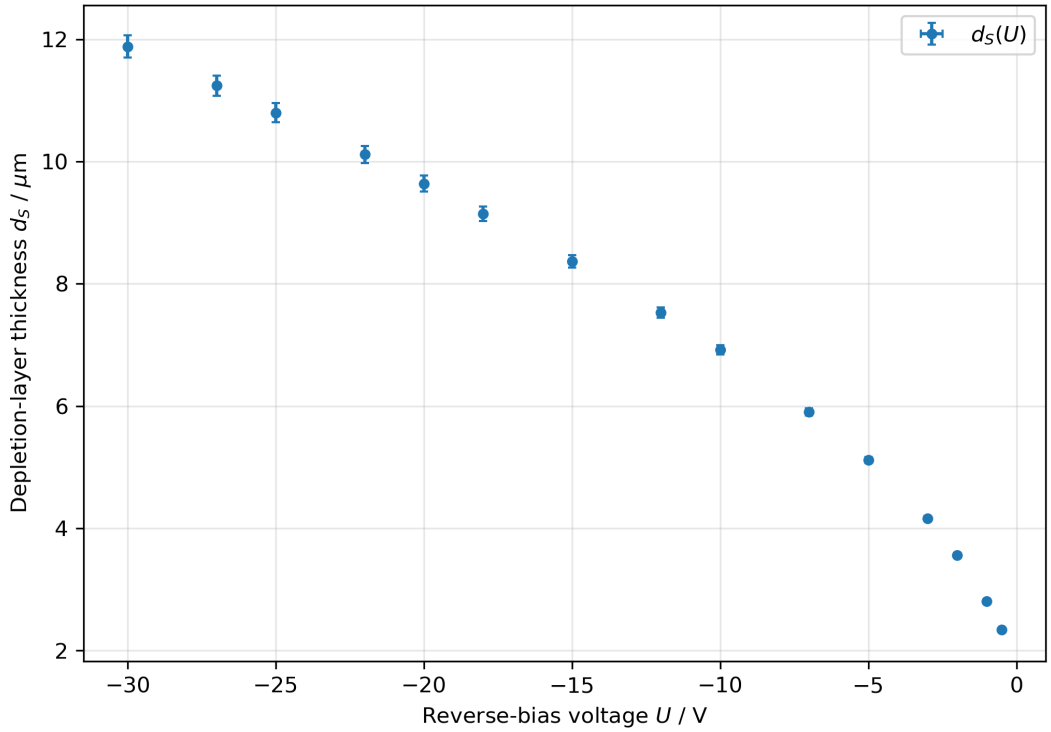


Figure 6: Depletion-layer thickness d_S as a function of reverse-bias voltage.

For vanishing applied voltage, $C_S = C_0$, so the zero-bias depletion-layer thickness is

$$d_0 = \frac{\varepsilon_r \varepsilon_0 A}{C_0}. \tag{127}$$

Using the fitted value of C_0, this gives

$$d_0 = 1.826\,\mu\text{m}. \tag{128}$$

Including the fit uncertainty and the systematic uncertainty of the inductance, the final result is

$$d_0 = 1.826 \pm 0.096\,\mu\text{m}. \tag{129}$$

6.7 Uncertainty Analysis

The resonance frequency uncertainty was estimated as

$$\Delta f = 0.05\,\text{kHz}. \tag{130}$$

The uncertainty in the total capacitance due to the frequency uncertainty follows from Eq. (115). Since

$$C_{\text{tot}} \propto f^{-2}, \tag{131}$$

the relative uncertainty is

$$\frac{\Delta C_{\text{tot}}}{C_{\text{tot}}} = 2\frac{\Delta f}{f}. \tag{132}$$

The capacitance error bars in Fig. 5 include this frequency uncertainty. In addition, the inductance has an unknown exact tolerance. Therefore, a systematic uncertainty of

$$\frac{\Delta L}{L} = 5\% \tag{133}$$

was assumed. Since

$$C_{\text{tot}} \propto \frac{1}{L}, \tag{134}$$

this uncertainty affects the absolute capacitance scale. It was treated separately from the frequency uncertainty because it shifts the capacitance values systematically rather than producing random point-to-point scatter.

For the zero-bias depletion-layer thickness,

$$d_0 = \frac{\varepsilon_r \varepsilon_0 A}{C_0}, \tag{135}$$

the relative uncertainty from the fitted capacitance is

$$\frac{\Delta d_0}{d_0} = \frac{\Delta C_0}{C_0}. \tag{136}$$

The fit uncertainty contribution was

$$\Delta d_{0,\mathrm{fit}} = 0.031\,\mu\mathrm{m}. \tag{137}$$

The systematic contribution from the assumed inductance tolerance was

$$\Delta d_{0,L} = 0.091\,\mu\mathrm{m}. \tag{138}$$

The total uncertainty was calculated by adding these independent contributions in quadrature:

$$\Delta d_0 = \sqrt{\Delta d_{0,\mathrm{fit}}^2 + \Delta d_{0,L}^2}. \tag{139}$$

This gives

$$\Delta d_0 = 0.096\,\mu\mathrm{m}. \tag{140}$$

6.8 Discussion

The measured capacitance decreases continuously as the magnitude of the reverse-bias voltage increases. This is the expected behaviour of a reverse-biased pn-junction. A larger reverse voltage widens the depletion region, and since the depletion region acts approximately like a parallel-plate capacitor, a larger depletion-layer thickness corresponds to a smaller capacitance.

The fitted parallel capacitance

$$C_G = 0.0708 \pm 0.0031\,\mathrm{nF} \tag{141}$$

is small compared with the zero-bias depletion capacitance but becomes more relevant at large reverse voltages, where the depletion-layer capacitance is smaller. Including this term is therefore important for describing the full measured capacitance-voltage curve.

The fitted diffusion voltage

$$U_D = 0.7382 \pm 0.0388\,\mathrm{V} \tag{142}$$

is of a reasonable order of magnitude for a silicon pn-junction. The reduced chi-square value close to 1 indicates that the fit model is consistent with the data and the assumed uncertainty in the resonance frequency.

The calculated depletion-layer thickness increases from approximately

$$d_S = 2.335\,\mu\mathrm{m} \tag{143}$$

at $-0.5\,\mathrm{V}$ to approximately

$$d_S = 11.885\,\mu\mathrm{m} \tag{144}$$

at $-30\,\mathrm{V}$. This confirms the expected widening of the depletion region under increasing reverse bias. The final value for the zero-bias depletion-layer thickness was

$$d_0 = 1.826 \pm 0.096\,\mu\mathrm{m}. \tag{145}$$

The dominant contribution to this uncertainty is the assumed systematic uncertainty of the inductance. Therefore, the relative voltage dependence of the capacitance and depletion-layer thickness is more precise than the absolute value of d_0.

7 Discussion

The measured current-voltage characteristics show the expected qualitative behaviour of the investigated semiconductor diodes. The Si diode and the Zener diode in forward bias both show the typical non-linear increase of current with increasing forward voltage. At low voltages, only a very small current flows, while at higher forward voltages the current rises rapidly. This behaviour is consistent with the exponential dependence predicted by the Shockley equation. The red LED also shows a non-linear forward characteristic, but the noticeable current increase occurs at a higher voltage than for the Si diode. This is expected because the LED threshold voltage is related to the photon energy of the emitted light.

The Si diode analysis gave an emission coefficient of

$$n = 1.901 \pm 0.003. \tag{146}$$

This value is larger than the ideal value $n = 1$, which indicates that the real diode does not behave like an ideal diffusion-limited pn-junction. Possible reasons include recombination effects in the depletion region, series resistance, non-ideal contact resistances, and deviations from the assumptions of the Shockley model. The result is nevertheless physically reasonable, since real diodes often show emission coefficients between 1 and 2 depending on the dominant current transport mechanism and the chosen fit region. The quality of the semi-logarithmic fit shows that the selected current interval was suitable for applying the linearized Shockley equation.

For the Zener diode, a clear breakdown region was observed in reverse bias. The breakdown voltage obtained from the linear extrapolation was

$$U_Z = 5.478 \pm 0.003\,\mathrm{V}. \tag{147}$$

Since the exact nominal Zener diode type was not known, a direct comparison with a specific nominal value was not possible. However, the measured value is of the correct order of magnitude for the Zener diodes listed for the experiment. The direct current resistance at $0.2I_{Z,\max}$ was found to be

$$R = 2642.2 \pm 64.8\,\Omega, \tag{148}$$

whereas the differential resistance in the fitted breakdown region was

$$r_d = 10.4 \pm 0.4\,\Omega. \tag{149}$$

The large difference between these two values is expected. The direct current resistance describes the ratio U/I at one operating point, while the differential resistance describes the local slope of the characteristic. In the breakdown region, the voltage changes only slightly even when the current changes significantly, which leads to a small differential resistance.

The threshold voltage of the red LED was determined as

$$U_S = 1.939 \pm 0.006\,\mathrm{V}. \tag{150}$$

Using the approximation $eU_S = hc/\lambda$, the average emitted wavelength was calculated as

$$\lambda = 639 \pm 2\,\mathrm{nm}. \tag{151}$$

This wavelength lies in the red region of the visible spectrum and agrees well with the observed colour of the LED. The agreement confirms that the threshold-voltage method gives a reasonable estimate of the emitted wavelength. However, the method is only approximate because an LED emits over a finite spectral range, and not all electrical energy is necessarily converted into photon energy. In addition, the extrapolated threshold voltage depends on the selected linear fit region of the forward characteristic.

The capacitance measurement of the Si power diode also gave physically consistent results. The total capacitance decreased from approximately

$$C_{\mathrm{tot}} = 1.1894\,\mathrm{nF} \tag{152}$$

at $-0.5\,\mathrm{V}$ to approximately

$$C_{\mathrm{tot}} = 0.2905\,\mathrm{nF} \tag{153}$$

at $-30\,\mathrm{V}$. This trend agrees with the expected behaviour of a reverse-biased pn-junction. As the magnitude of the reverse-bias voltage increases, the depletion region widens. Since the depletion region behaves approximately like a parallel-plate capacitor, the capacitance decreases as the depletion-layer thickness increases.

The non-linear fit of the capacitance-voltage curve gave

$$C_G = 0.0708 \pm 0.0031\,\mathrm{nF}, \tag{154}$$

$$C_0 = 1.4306 \pm 0.0242\,\mathrm{nF}, \tag{155}$$

and

$$U_D = 0.7382 \pm 0.0388\,\mathrm{V}. \tag{156}$$

The fitted diffusion voltage is reasonable for a silicon junction. The reduced chi-square value

$$\chi^2_{\mathrm{red}} = 0.975 \tag{157}$$

shows that the fit describes the measured capacitance data well within the assumed resonance-frequency uncertainty. The fitted parallel capacitance C_G is small compared with the zero-bias depletion capacitance, but it becomes increasingly important at larger reverse voltages, where the depletion capacitance is smaller. Including C_G is therefore necessary for a good description of the measured data.

From the fitted zero-bias capacitance, the depletion-layer thickness for vanishing applied voltage was determined as

$$d_0 = 1.826 \pm 0.096\,\mu\mathrm{m}. \tag{158}$$

The calculated voltage-dependent depletion-layer thickness increased from approximately $2.335\,\mu$m at $-0.5\,$V to $11.885\,\mu$m at $-30\,$V. This confirms the expected widening of the depletion layer under increasing reverse bias.

The main sources of uncertainty in the experiment were the estimated voltage and current measurement uncertainties, the choice of fit intervals, the determination of the resonance condition from the Lissajous figure, and the uncertainty of the inductance used in the resonance circuit. For Task 2, the fit interval selection was especially important, since the diode characteristics are only approximately linear in the chosen transformed or high-current regions. For Task 3, the point-to-point scatter was small, but the assumed 5% systematic uncertainty of the inductance affected the absolute capacitance scale and therefore the final value of d_0. This means that the relative voltage dependence of the capacitance is more reliable than the absolute capacitance scale.

Overall, the results of all three tasks are consistent with the expected physical behaviour of pn-junction diodes. The current-voltage measurements demonstrate rectification, breakdown, and light emission, while the resonance measurement shows the voltage dependence of the depletion-layer capacitance. The extracted parameters are physically plausible and agree with the qualitative predictions of semiconductor diode theory within the estimated uncertainties.

8 Conclusion

In this experiment, the current-voltage characteristics of a Si diode, a Zener diode, and a red LED were measured and analysed. The Si diode and the Zener diode in forward bias showed the expected non-linear increase of current with voltage, while the Zener diode in reverse bias showed a clear breakdown region. The red LED required a higher forward voltage before a significant current increase occurred, consistent with its light-emitting behaviour.

From the semi-logarithmic analysis of the Si diode characteristic, the emission coefficient was determined as

$$n = 1.901 \pm 0.003. \tag{159}$$

For the Zener diode, the breakdown voltage was found to be

$$U_Z = 5.478 \pm 0.003\,\mathrm{V}. \tag{160}$$

At $0.2I_{Z,\mathrm{max}}$, the direct current resistance was

$$R = 2642.2 \pm 64.8\,\Omega, \tag{161}$$

and the differential resistance was

$$r_d = 10.4 \pm 0.4\,\Omega. \tag{162}$$

For the red LED, the threshold voltage was determined as

$$U_S = 1.939 \pm 0.006\,\mathrm{V}, \tag{163}$$

which corresponds to an estimated average wavelength of

$$\lambda = 639 \pm 2\,\mathrm{nm}. \tag{164}$$

This value is consistent with the observed red colour of the LED.

The junction capacitance of a Si power diode was measured using a resonance method. The capacitance decreased with increasing reverse-bias voltage, which agrees with the expected widening of the depletion layer. A non-linear fit of the capacitance-voltage characteristic gave

$$C_G = 0.0708 \pm 0.0031\,\mathrm{nF}, \tag{165}$$

$$C_0 = 1.4306 \pm 0.0242\,\mathrm{nF}, \tag{166}$$

and

$$U_D = 0.7382 \pm 0.0388\,\mathrm{V}. \tag{167}$$

The zero-bias depletion-layer thickness was calculated as

$$d_0 = 1.826 \pm 0.096\,\mu\mathrm{m}. \tag{168}$$

Overall, the measured diode characteristics and extracted parameters agree well with the expected behaviour of pn-junction diodes. The main limitations were the estimated instrument uncertainties, the choice of fit intervals, and the systematic uncertainty of the inductance in the resonance measurement.

9 References

M. Ziese, E14e Semiconductor Diodes, University of Leipzig, June 2026.